

System and Network Security

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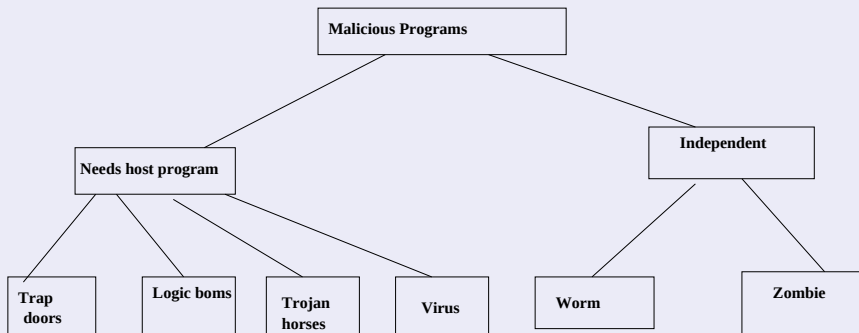
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MALICIOUS SOFTWARE

Taxonomy of Malicious programs



● Trap Doors

- ▶ A trap door is a secret entry point into a program that allows someone that is aware of the trapdoor to gain access without going the usual security access procedures.

● Logic Bomb

- ▶ One of the oldest types of program threat, predating viruses and worms, is the logic bomb.
- ▶ It is code embedded in some legitimate program that is set to explode when certain conditions are met.
For example, certain conditions may be a particular day of the week or date, or a particular user running the application.

● Trojan Horses

- ▶ A Trojan horse is a useful, or apparently useful, program or command procedure containing hidden code that, when invoked, performs some useful unwanted or harmful function.

● **Zombie**

- ▶ A zombie is a program that secretly takes over another Internet-attached computer and then uses that computer to launch attacks that are difficult to trace to the Zombie's creator.
- ▶ These are useful for denial-of-service attacks, typically against targeted Web sites.

● **Viruses**

- ▶ A virus is a program that can infect other programs by modifying them.
- ▶ The modification may include a copy of the virus program, which can then go on to infect other programs.

● **Worms**

- ▶ A worm propagates itself from system to system.
- ▶ A worm actively seeks out more machines to infect and each machine that is infected serves as an automated launching pad for attacks on other machines.

- **Ransomware:** It is considered as a form of malware that has ability to encrypt a victim's files containing the sensitive and confidential information. The attacker then requests a ransom from the victim in order to reinstate access to the data based on satisfactory payment.
- **Adware:** It is treated as unwanted software that is developed in order to throw various advertisements up on the victim's screen (most often within a web browser). Adware helps the attacker in generating revenue for the developers maintained by an attacker by “involuntarily displaying online advertisements in the user interface of the software or on a screen that pops up in the user's face during the installation process”.
- **Spyware:** It is treated as “unwanted software that penetrates a victim's computing device, steals the victim's internet usage data and sensitive information”.

- A virus can do anything that other programs do. The only difference is that it attaches itself to another program and executes secretly when the host program is run. Once a virus is executing, it can perform any function, such as erasing files and programs.
- During its lifetime, a typical virus goes through the following four phases:
 - ▶ **Dormant phase:** The virus is idle in this phase.
 - ▶ **Propagation phase:** The virus places an identical copy of itself into other programs or into certain system areas on the disk. Each infected program contains a clone of the virus, which itself enters a propagation phase.
 - ▶ **Triggering phase:** The virus is activated to perform the function for which it was intended. This phase can be caused by a variety of system events, including a count of the number of times that this copy of the virus has made copies of itself.
 - ▶ **Execution phase:** This phase may be harmless, such as a message on the screen, or damaging, such as the destruction of programs and data files.

Virus Structure

- A virus can be prepended or postpended to an executable program, or it can be embedded in some other fashion.
- The key to its operation is that the infected program, when invoked, will first execute the virus code and then execute the original code of the program.
- The structure of a virus can be either simple or compressed.

A Simple Virus

```
Program V:=
{ goto main;
1234567;
subroutine infect-executable :=
{ loop:
  file := get-random-executable-file;
  if (first-line-of-file == 1234567)
    then goto loop;
  else prepend V to file;
}
subroutine do-damage :=
{  whatever damage is to be done;
}

subroutine trigger-pulled: =
{
  if some condition holds then
    return true;
}

main:  main-program :=
      {infect-executable;
      if (trigger-pulled) then
        do-damage;
      goto next;
      }

next:
}
```

Logic for a Compression Virus

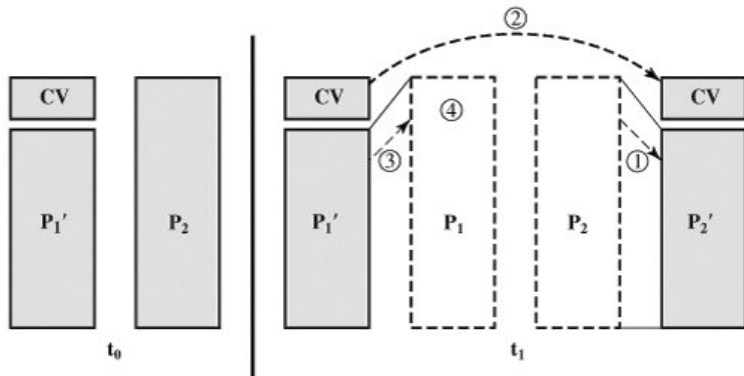
```
program CV: =  
{ goto main;  
  01234567;  
  
  subroutine infect-executable :=  
  { loop:  
    file := get-random-executable-file;  
    if (first-line-of-file == 01234567) then goto loop;  
    (1) compress file;  
    (2) prepend CV to file;  
  }  
  
  main: main-program :=  
  { if ask-permission then infect-executable;  
    (3) uncompress rest-of-file;  
    (4) run uncompressed file;  
  }  
}
```

Logic for a Compression Virus (Continued...)

- Assume that program P_1 is infected (denoted by P'_1) with the virus CV. When the program is invoked, control passes to its virus, which performs the following steps:
 - ▶ **Step 1:** For each uninfected file P_2 that is found, the virus first compresses that file to produce P'_2 , which is shorter than the original program by the size of the virus.
 - ▶ **Step 2:** A copy of the virus is prepended to the compressed program.
 - ▶ **Step 3:** The compressed version of the original infected program P'_1 , is uncompressed.
 - ▶ **Step 4:** The uncompressed original program is executed.

Note that in this example, the virus does nothing other than propagate. However, the previous example explains that the virus may include a logic bomb.

Example of a Compression Virus



- **Parasitic virus:** The traditional and still most common form of virus.

A parasitic virus attaches itself to executable files and replicates, when the infected program is executed, by finding other executable files to infect.

- **Memory-resident virus:** Lodges in main memory as part of a resident system program.

From that point on, the virus infects every program that executes.

- **Boot sector virus:** Infects a master boot record or boot record and spreads when a system is booted from the disk containing the virus.
- **Stealth virus:** A form of virus explicitly designed to hide itself from detection by anti-virus software.
- **Polymorphic virus:** A virus that mutates with every infection, making detection by the “signature” of the virus impossible.

Macro viruses are particularly threatening for a number of reasons:

- A macro virus is platform independent.
For example, virtually all of the macro viruses infect Microsoft Word documents. Any hardware platform and operating system that supports Word can be infected.
- Macro viruses infect documents, not executable portions of code.
- Macro viruses are easily spread. A very common method is by electronic mail (e-mail).

A more recent development in malicious software is the e-mail virus.

- The e-mail virus sends itself to everyone on the mailing list in the user's e-mail package.
- The virus does local damage.
- The virus propagates itself as soon as activated (either by opening an e-mail attachment or by opening the e-mail) to all of the e-mail addresses known to the infected host.

Network worm programs use network connections to spread from system to system.

To replicate itself, a network worm uses some sort of network vehicle. Examples include

- **Electronic mail facility:** A worm mails a copy of itself to other systems.
- **Remote execution capability:** A worm executes a copy of itself on another system.
- **Remote login capability:** A worm logs onto a remote system as a user and then uses commands to copy itself from system to the other.

A network worm exhibits the same characteristics as a computer virus. It has the phases: a dormant phase, a propagation phase, a triggering phase, and an execution phase.

The propagation phase generally performs the following functions:

- Search for other systems to infect by examining host tables or similar repositories of remote system addresses.
- Establish a connection with a remote system.
- Copy itself to the remote system and cause the copy to be run.

- **Detection:** Once the infection has occurred, determine that it has occurred and locate the virus.
- **Identification:** Once detection has been achieved, identify the specific virus that has infected a program.
- **Removal:** Once the specific virus has been identified, remove all traces of the virus from the infected program and restore it to its original state.
Remove the virus from all infected systems so that disease cannot spread further.

Four generations of anti-virus software:

- **First generation:** A first-generation scanner requires a virus signature to identify a virus.
Another type of first-generation scanner maintains a record of the length of programs and looks for changes in length.
- **Second generation:** A second-generation scanner does not rely on a specific signature. It uses heuristic rules to search for portable virus infection.

Example: A scanner may look for the beginning of an encryption loop used in a polymorphic virus and discover the encryption key. Once the key is discovered, the scanner can decrypt the virus to identify it, and then remove the infection and return the program to service.

- **Third generation:** Third-generation programs are memory resident. They identify a virus by its actions rather than its structure in an infected program.
Such programs have the advantage that it is not necessary to develop signatures and heuristics for a wide array of viruses.
- **Fourth generation:** Fourth-generation products are packages consisting of a variety of anti-virus techniques used in conjunction. These include scanning and activity trap components.

William Stallings, "Cryptography and Network Security: Theory and Practices", Pearson Education Asia, 2010.

All the best!!!